📓 Super Easy C++ Notes (Hinglish + Clear + With Output)

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1) Inline Function

☞ **Definition:** Inline function ka matlab hota hai ki function ka code seedha copy ho jata hai call wali jagah pe. Isse program fast hota hai.

Code:

```
#include <iostream>
using namespace std;

inline int square(int x) {
    return x * x;
}

int main() {
    cout << square(5);
}
```

Output:

```
25
```

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2) Recursive Function

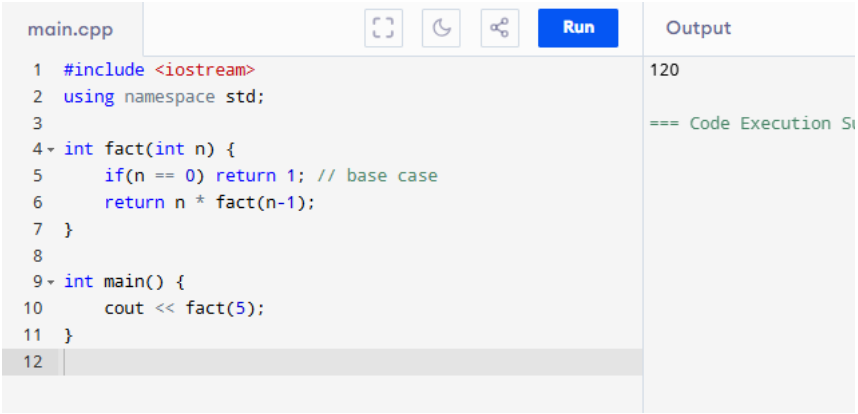
☞ **Definition:** Recursive function apne aap ko call karta hai. Base case hona zaroori hai.

Code:

```
#include <iostream>
using namespace std;

int fact(int n) {
    if(n == 0) return 1; // base case
    return n * fact(n-1);
}

int main() {
    cout << fact(5);
}
```



Output:

```
120
```

CODE → CLICK ME

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3) Function Pointer

☞ **Definition:** Function pointer ek pointer hai jo function ka address store karta hai.

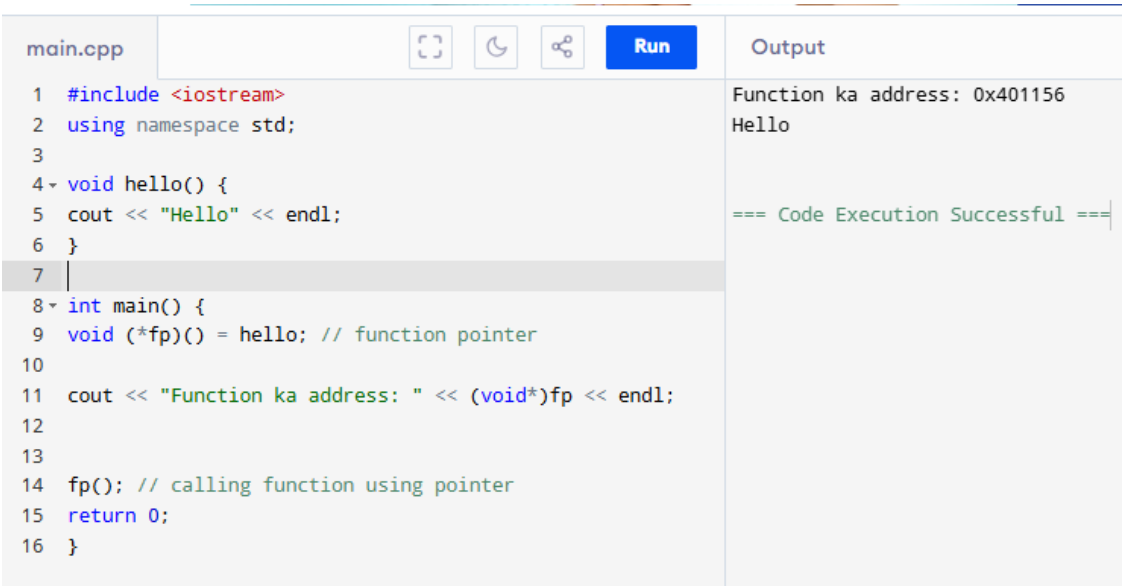
Code:

```
#include <iostream>
using namespace std;

void hello() { cout << "Hello"; }

int main() {
    void (*fp)() = hello; // function pointer
    fp();
}
```

Output:

```
Hello
```

ANOTHER ➔



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4) Multi-Dimensional Array

☞ **Definition:** Multi-D array ka matlab hota hai matrix type data (rows × cols).

Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[2][2] = {{1,2},{3,4}};
    for(int i=0;i<2;i++){
        for(int j=0;j<2;j++){
            cout<<arr[i][j]<<" ";
        }
        cout<<endl;
    }
}
```

Output:

```
1 2
3 4
```

CLICK - > CODE

5) String in C++

👉 **Definition:** String ek special class hai jo text store aur modify karne ke liye use hoti hai.

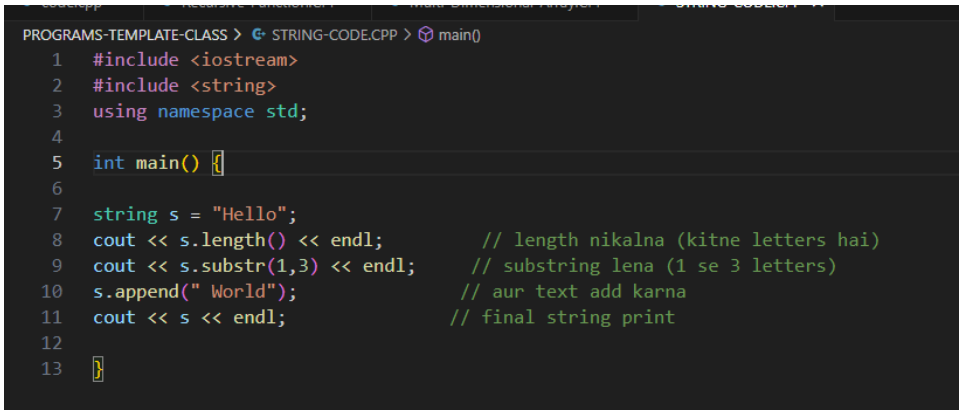
Code:

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string s = "Hello";
    cout << s.length() << endl;  // length
    cout << s.substr(1,3) << endl; // substring
    s.append(" World");
    cout << s << endl;
    return 0;
}
```

Output:

```
5
ell
Hello World
```



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6) Goto Statement

☞ Definition:

Goto ek jump statement hai jo seedha label par le jata hai. Use karna recommend nahi hai.

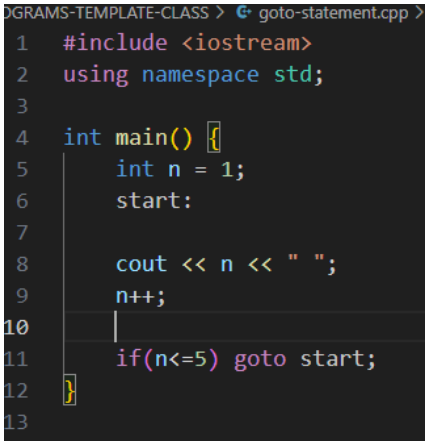
Code:

```
#include <iostream>
using namespace std;

int main() {
    int n = 1;
    start:
    cout << n << " ";
    n++;
    if(n<=5) goto start;
}
```

Output:

```
1 2 3 4 5
```



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7) Templates

☞ **Definition:** Template se ek hi code different data types ke liye kaam karta hai.

(a) Function Template

```
#include <iostream>
using namespace std;

template <typename T>

T add(T a, T b) {
    return a+b;
}

int main() {
    cout << add(5,6) << endl;
    cout << add(2.5,3.5) << endl;
}
```

Output:

```
11
6
```

Code -> [Click me]

(b) Class Template

✅ Template in C++ (Definition)

👉 Template ek **blueprint** hota hai jisme hum **generic code** likhte hai, jo alag-alag data types (int, float, double, string…) ke liye automatically kaam karta hai **without code duplicate**.

PROGRAMS-TEMPLATE-CLASS > class-template-1.cpp > ...

```cpp
#include <iostream>
using namespace std;

template <class T>
T add(T a, T b) {
    return a + b;
}

int main() {
    cout << add(5, 10) << endl;      // int
    cout << add(2.5, 3.5) << endl;  // double
    cout << add(string("Hi "), string("Bhai")) << endl; // string
}

```

Code—> click me

Output:

```
10
```